$$g''\left(\ln(2)\cdot\frac{m}{n}\right)<0$$

We'll spare this step for the sake of space, but you can double check that it's indeed true.

It's worth noting that

- The formula for `k` gives us a single, exact value for the optimal choice of the number of hash functions;
- `k` must obviously be an integer, so the result needs to be rounded;
- Larger values for `k` mean worst performance for insert and look-up (because more hash functions need to be computed for each element), so a compromise with slightly smaller values of `k` can be preferred.

If we use the best value for `k` as computed above, this means that the rate of false positives `f` becomes

$$f=\left(\frac{1}{2}\right)^k=(0.6185)^{\frac{m}{n}}$$

By replacing the value `k` in the formula for `p(n,m,k)`, we can get a new formula that links the number of storage bits to the (maximum) number of elements that can be stored, independently of `k` (the value for `k` can be computed later) to guarantee a false-positive probability smaller than a certain value `p`: